

Nuzefatide — Bear Pitch (2026-08-24)

THE CASE AGAINST NUZEFATIDE — a biology bear pitch with freezable predictions

2026-08-24 · principal-directed (“I am totally ok to make a prediction against the biology of nuzefatide”) · informs the deferred BCYC entry (instr #100), rules nothing

The pitch in one paragraph. Bicycle Therapeutics has redirected its burn onto nuzefatide (BT5528), an EphA2-targeted toxin conjugate. EphA2 is a four-time clinical graveyard — a fact the sponsor’s own 2026 review concedes by name. The drug’s only published efficacy number, from its own Phase 1 in *JCO* (Sept 2024), is a **6.7% response rate at the selected go-forward dose** — chemo-noise level — while the format’s pharmacology inverts its sales pitch: the targeting molecule survives **under an hour** in circulation but the free toxin it releases circulates for **two days**. The company then chose, as its pivot indication, the one tumor type (2L+ pancreatic) where single-digit response benchmarks make a weak drug spinnable — with a 39-patient trial that cannot be falsified until 2029. We predict the imminent final-data disclosure does not rescue the asset, and we say so with frozen numbers.

1 · The EphA2 graveyard (the prior)

Every prior clinical program against this target has been halted — **the sponsor’s own review says so verbatim**: “efficacy and safety issues have halted clinical development of previous EphA2-targeting agents including MEDI-547, DS-8895a, MM-310, and dasatinib” (Bennett et al. — BicycleTx authors — *J Transl Med* 2026, PMID 41782030). - **MEDI-547** (MedImmune, antibody-ADC, auristatin): Phase 1 **stopped at the first dose level** — six patients at 0.08 mg/kg, all discontinued; treatment-related bleeding and coagulation events (hemorrhage n=3, epistaxis n=2); “lower doses were not explored and an MTD could not be selected” (*Invest New Drugs* 2013, PMID 22370972). The target sits on normal vasculature; hitting it with a long-circulating toxin was a disaster. - **DS-8895a** (Daiichi, afucosylated ADCC antibody): two Phase 1s (PMID 31412935, 35404015) — safety and imaging established, no efficacy signal worth developing; discontinued. - **MM-310** (EphA2-targeted docetaxel nanoliposome) and **dasatinib** (EphA2 among off-targets): halted (per the sponsor review; MM-310’s own literature trail

is thin — flagged, not embellished). **Base rate: 0-for-4.** No EphA2-directed therapeutic has ever cleared Phase 2. Any new EphA2 conjugate starts from a single-digit prior before its own data speaks.

2 · Nuzefatide’s own number — published, and worse than “undisclosed”

The desk’s earlier dive recorded the pivot asset’s efficacy as “a characterization, not a number.” **Correction: the number exists.** The first-in-human dose-escalation was published in *JCO* (2024;42:3443-52, PMID 39231383, n=45): - **At the selected RP2D (6.5 mg/m² q2w): ORR 6.7%, disease-control rate 20.0%.** - Tox: manageable (G_≥3 neutropenia 22.2%; no MEDI-547-style bleeding) — the format genuinely retired the class safety failure. Dosing reached 10 mg/m² vs MEDI-547’s death at first dose. - **PK, from the same paper: BT5528 half-life 0.4–0.7 h; released MMAE half-life 35–47 h.** The tumor sees the targeting vector for minutes; the body sees free payload for days. The G_≥3 neutropenia is the signature of exactly that — systemic MMAE without the targeting. The “no headline ORR after seven years” pattern is now explained: there is a number, and it does not get headlined. The undisclosed remainder (the ~240-patient expansion cohorts, PCD passed 7/31/26) is what the imminent disclosure covers — see the prediction.

3 · The format-vs-indication contradiction

Bicycle’s core claim: small size → fast, deep tumor penetration; renal clearance → brief exposure → safety. The mirror: **brief exposure must still deliver enough payload**, and the chosen pivot indication is metastatic **pancreatic ductal adenocarcinoma — the most stroma-dense, hypoperfused major solid tumor**. A sub-hour delivery window into the least-perfused tumor class is a mechanistic bet *against* the indication’s defining biology. Honest mirror check, conceded: EphA2 target expression in PDAC is real (abundant by IHC, Giaginis 2010, PMID 19949912; the company shows 16/16 PDX expression and human Ga-68 imaging engagement) — the target is present; the question is whether a transient vector at 8 mg/m² q2w can deliver a cytotoxic dose through PDAC stroma when it managed 6.7% ORR in better-perfused tumors.

4 · Indication-choice forensics

With ~240 expansion slots dosed across urothelial, ovarian, NSCLC, HNSCC and TNBC since 2019 — through at least four ASCO/ESMO cycles — no expansion cohort’s response rate has ever led a press release. The sole company characterization is “promising anti-tumor activity” on an n=14 mUC+nivo cohort. This company press-releases zelenectide data constantly; **selective silence is selection**. Then the pivot trial (NCT07450859) chose second-line-plus PDAC: n=39, single-arm, ORR primary, first dosed April 2026, primary completion **March 2029**. In 2L+ PDAC, chemotherapy benchmarks run single digits to mid-teens — it is the only indication where a 10–15% ORR reads as a win. The structure is bar-lowering plus falsifiability-deferral: three fundable years before the asset can fail.

5 · The revealed-preference chain

March 2026: the company **killed the asset with randomized data** (zelenectide: 58–62% confirmed 1L BICR ORR) after adverse regulatory feedback — and pivoted the firm onto the asset with a published 6.7% RP2D ORR and an unfalsifiable-until-2029 lead trial. Zero insider buys on record. A strategic process five months without closure. The uncharitable-but-arguable read stands: **the least falsifiable asset is the most financeable one**, and the burn (~\$50M/qtr against \$510M) now runs through it.

6 · The honest counter-case (what would make the bear wrong)

- The tox reboot is real and unprecedented for this target — “safe at active-class doses” is an achievement no prior EphA2 program managed; safety alone has partnering value in radioligand form.
- Escalation-stage ORR **understates** enriched expansion cohorts as a rule; the 6.7% is $n \approx 15$ at one dose in mixed tumors; the undisclosed $n=14$ mUC cohort runs **WITH** nivolumab (combo uplift possible).
- EphA2 expression in PDAC is verified (16/16 PDX, human imaging); an 8 mg/m² dose is above the RP2D used for the 6.7%.
- The platform carries \$240M of cumulative partner inflows (Bayer/Novartis/Ionis/Genentech) — outside money keeps validating the *binders*, whatever the toxin-conjugate results say. **Falsifiers, stated in advance:** a disclosed confirmed ORR $\geq 20\%$ ($n \geq 20$) in any BT5528-100 cohort; or ≥ 3 responses in the first ~ 15 PDAC patients at any interim. Either kills this pitch and triggers the entry court instead.

7 · FREEZABLE PREDICTIONS (proposed specs — desk freezes only after principal review)

1. **NUZE-DATA | cat_date 2027-03-31 | our_p = 0.12 (i.e., 88% the bear holds)**
FAVORABLE(bear) = the complete BT5528-100 (NCT04180371) final/expansion data disclosure — press release, conference presentation, or filing, whichever first — does **NOT** report a confirmed ORR $\geq 20\%$ with $n \geq 20$ in any single cohort (monotherapy or combination) by the cat_date. *Arithmetic:* $P(\text{true enriched-cohort ORR} \geq 20\% \mid \text{RP2D escalation ORR } 6.7\%) \approx 0.20\text{--}0.25$ (enrichment
◦ combo uplift, same dose class) $\times P(\text{still undisclosed through } \geq 4 \text{ conference cycles} \mid \text{that good}) \approx 0.4 \rightarrow \approx 0.09$; rounded up to 0.12 for the unobserved nivo-combo cohort. External anchor: none — no sell-side estimate for this readout located (orphan datum).
2. **NUZE-PDAC | cat_date 2029-09-30 | our_p = 0.15** FAVORABLE(bear) = NCT07450859 does **NOT** meet its ORR primary endpoint per its own analysis plan (grade on the company’s topline release; if the SAP bar is never disclosed, grade against confirmed ORR $\geq 15\%$, the low end of what a single-arm $n=39$ registration-signal design must clear). *Arithmetic:* class base rate 0-for-4 $\times$ published same-drug 6.7% at RP2D in easier-perfused tumors $\times$ stroma mismatch, offset by the higher 8 mg/m² dose, PDAC target verification, and a possibly-low SAP bar $\rightarrow 0.15$. SAP is UNVERIFIED — flagged inside the spec. (*Both are bear-affirmative framings: high*

p = the drug fails. The desk's Brier discipline reads direction from the resolution bar, not the label.)

8 · File implications (recommendation, not a ruling)

- **Correct the record:** the dive's "no disclosed ORR anywhere" is superseded — JCO 2024 published 6.7%/20% at the RP2D. This *strengthens* the deferral logic materially.
- **Raise the deferred-entry bar to a closure rule:** if the imminent disclosure shows no cohort at confirmed ORR $\geq 20\%$ ($n \geq 20$), **close the BCYC entry file** — do not keep deferring. A 6.7%-ORR pivot asset walking toward a 2029 PDAC readout, funded by a melting cash pile, is not a deferral candidate; the zeke-2H26 zero-carry watch can survive as a standalone sensor without any position.
- A short expression would require the six-gate short court (ADV \$1.4M/day and borrow economics likely fail it); the frozen calls above are the honest instrument for the principal's conviction.

What could go wrong with THIS pitch (pre-mortem on the bear)

1. *Tripwire-less (disclosure-selection inversion):* the company may have held expansion data for a partnering data room rather than because it is weak — silence read as weakness when it was deal-process discipline. No sensor distinguishes these before the disclosure.
2. *The combo confound:* the $n=14$ cohort runs with nivolumab; a strong combo number would be spinnable as validation even though it tests a different hypothesis than the monotherapy pivot.
3. *Reflexive leg:* the desk holds a deferred LONG instruction on BCYC — a bear pitch that over-anchors us could cause skipping a genuinely repriced entry if the disclosure surprises. **"If this pitch costs us money, the most likely reason will be that expansion enrichment did what escalation never could, the disclosure printed a 25%+ combo ORR, and we had pre-committed ourselves to reading silence as failure."**

Sources: JCO 2024;42:3443 (PMID 39231383) · Invest New Drugs 2013 (PMID 22370972) · J Transl Med 2026 (PMID 41782030) · Invest New Drugs 2022 (PMID 35404015) · J Immunother Cancer 2019 (PMID 31412935) · Pathol Oncol Res 2010 (PMID 19949912) · ClinicalTrials.gov API v2: NCT04180371, NCT07450859, NCT07681674 (pulled 2026-08-24) · BCYC 10-Q via desk dive 2026-08-24. UNVERIFIED: MM-310's primary halt documentation (sponsor-review sourced only); the $n=14$ combo cohort ORR; the PDAC SAP bar; the regulatory-feedback content behind the zelenectide demotion.